Figure 1: Key applications of deep learning

Pre-trained models for multiple research domains

Pre-trained models are used to implement deep learning rapidly with high accuracy (see Table 2). These models have weights, which can be imported by researchers and scientists to deploy the deep learning application quickly in a particular domain without modelling from scratch.

Table 2: Prominent pre-trained deep learning models for assorted applications

Object detection and image analytics	• Xception • VGG16 • VGG19 • ResNet • ResNetV2 • ResNeXt • InceptionV3 • InceptionResNetV2 • MobileNet • MobileNetV2 • DenseNet • NASNet • YOLO
Natural language processing	• OpenAI GPT-3 • Google BERT • Google ALBERT • Google Transformer-XL • ULMFiT • Facebook RoBERTa • Microsoft CodeBERT • ELMo • XLNet
Audio and speech	• Wavenet • Lip Reading • MusicGenreClassification • Audioset • DeepSpeech • Waveglow • Loop • TTS • ESPNET • MXNET-Audio

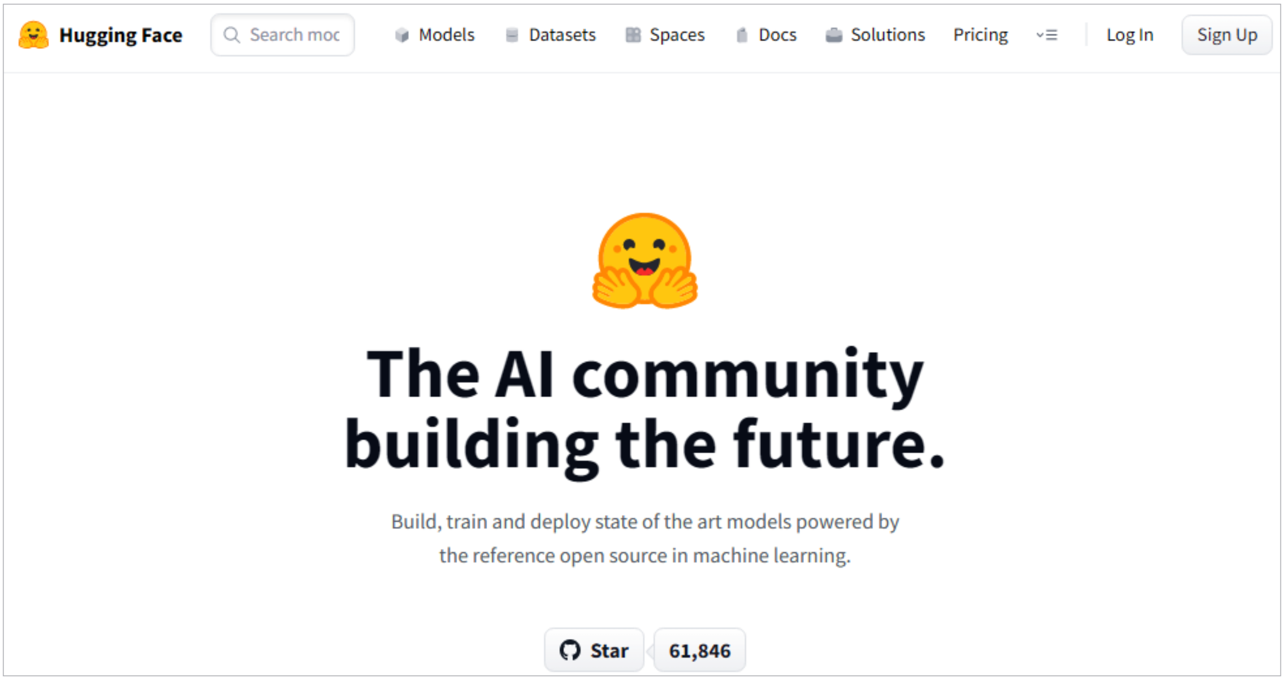

Figure 2: Online platform *huggingface.co* for pre-trained models